

MATERIAL SAFETY DATA SHEET (MSDS)

Standard GHS Compliance Document for Laboratory Safety

SECTION 1: CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Chemical Name	Ethanol, Absolute
Chemical Formula	C2H5OH
CAS Number	64-17-5
Document Purpose	Academic RAG Validation Dataset

SECTION 2: HAZARDS IDENTIFICATION

 Highly flammable liquid and vapor. Causes serious eye irritation. May cause drowsiness or dizziness.

SECTION 4: FIRST AID MEASURES

- Inhalation:** Move to fresh air. If symptoms persist, obtain medical attention.
- Skin Contact:** Wash off with soap and plenty of water. Remove contaminated garments.
- Eye Contact:** Thoroughly rinse with plenty of water for at least 15 minutes and consult a doctor if irritation persists.
- Ingestion:** Rinse mouth with water. Do not induce vomiting without medical advice. Never give anything by mouth to an unconscious person.

SECTION 6: ACCIDENTAL RELEASE MEASURES

- Precautions:** Eliminate all sources of ignition (no smoking, flares, sparks or flames). Use spark-proof tools and explosion-proof equipment. Ensure adequate ventilation.
- Containment:** Contain spillage, soak up with non-combustible absorbent material (e.g. sand, earth, diatomaceous earth) and transfer to a container for disposal according to regulations.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Property	Value
Appearance	Clear, colorless liquid
Odor	Alcohol-like
pH	Neutral (7.0)
Boiling Point	78.3 °C (172.9 °F)
Melting Point	-114.1 °C (-173.4 °F)
Flash Point	13 °C (55.4 °F) - Closed cup
Density	0.789 g/cm³

SECTION 10: STABILITY AND REACTIVITY

Vapors may form explosive mixture with air. Incompatible with strong oxidizing agents, peroxides, acids, acid chlorides, and alkali metals.

SECTION 16: OTHER INFORMATION

This document is synthetically generated for testing Retrieval-Augmented Generation (RAG) system pipelines within clinical biochemistry and safety management training workflows.